

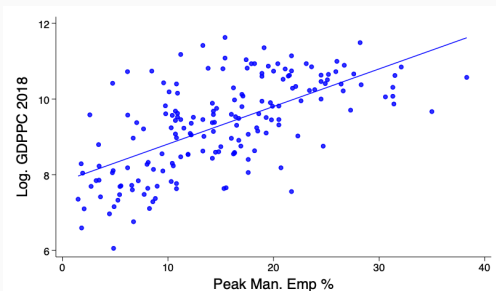
From Agriculture to Services: Development without Industrialization

Tancredi Rapone¹

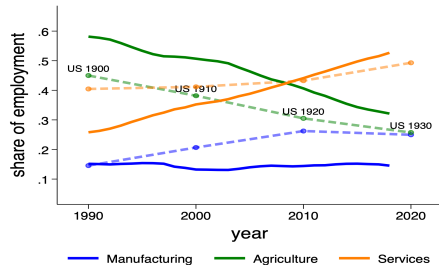
EWMEs, December 2025

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Motivation



(a) Manufacturing and GDPPC



(b) Emp. Shares over Time

Figure 1: Scatterplot of 2019 GDPPC against peak employment share in manufacturing (left) and sector shares in 42 developing countries (right). Data Sources: Groningen Growth and Development Center (10 Sector database and Economic Transformation database).

► Table

Q: Can developing countries close the income gap without manufacturing? Should governments implement industrial policy to foster industrial expansion?

Roadmap:

- Document three stylized facts:
 - ⇒ Countries with higher manufacturing peaks $\Rightarrow$ faster growth, better education, and quicker demographic transitions.
- Build a unified growth model to understand the implications of bypassing manufacturing on future growth possibilities.
- Calibrate to the US and change a preference shifter to simulate transitions with different peak manufacturing emp. shares.

Literature

Literature Review

I am contributing to several literatures:

- **Structural Change and Development:** Huneus and Rogerson (2023); Fujiwara and Matsuyama (2024); Sposi, Yi and Zhang (2021); Buera, Kaboski, Mestieri and O'Connell (2024), Kruse, Mensah and de Vries (2022); Rodrik (2016); Duarte and Restuccia (2010); Syrquin Chenery (1989); Kuznets (1959).
- ⇒ Add UGT to understand the consequences of premature de-industrialization on cross country income inequality.
- **Industrial Policy:** Juhász, Lane and Rodrik (2023); Chang and Andreoni (2020); Aghion, Cai, Dewatripont, Du, Harrison and Legros (2015); Rodrik (2008); Herrendorf, Rogerson and Valentinyi (2022).
- ⇒ Integrate structural change in a growth model.
- **UGT:** Galor and Weil (2000); Voigtlander and Voth (2013); Doepke (2004); Hansen and Prescott (2002); Parente and Prescott (2004).
- ⇒ Incorporate sectoral differences in models of UGT.

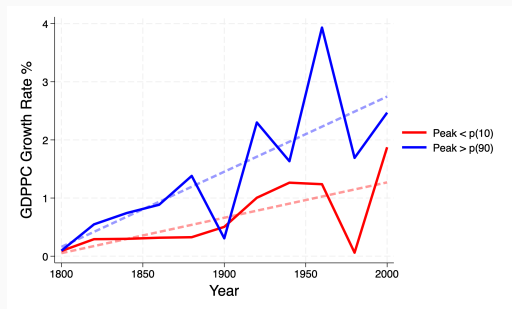
Data Sources

Data Sources

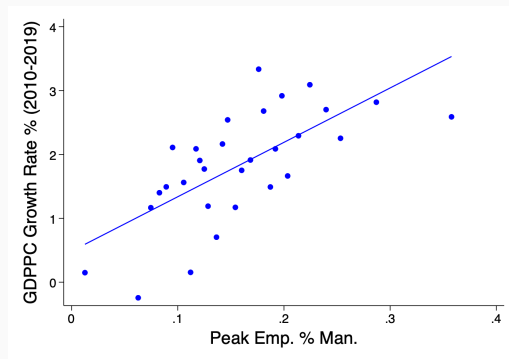
- **Sectoral employment shares:**
 - **Groningen Growth and Development Center (GGDC):**
 - Economic Transformation Database covers 41 low and middle income countries over 1990-2018.
 - 10 Sector Database covers 33 middle/high/low income countries over 1960 - 2010.
 - **ILO via Gapminder:** Remaining countries not covered by GGDC are filled in using Gapminder's data on sectoral employment shares.
 - **US Census of Population:** microdata covering the universe of population with information on employment, occupation and industry at the county level over 1850-present.
- **Fertility:**
 - **IPUMS International:** microdata on fertility and employment sector for 211 country/year samples.
 - **UN via Gapminder:** aggregate fertility rates 190 countries 1800-present.
- **Education:**
 - **Barro and Lee:** 2021 update provides data on years of schooling and human capital quality for 80 high-low and middle income countries over 1960-2020.

Stylized Facts

Stylized Fact 1: GDP p.c. Growth Rates and Manufacturing



(a) Over time



(b) Across countries

Figure 2: (left) Average GDP per capita growth rate over 1800-2020 grouped by top 10% peak emp. share in manufacturing and bottom 10%, for this figure I remove major oil exporters; (right) scatterplot of average GDP per capita growth rates over 2010-2020 against the peak employment share of manufacturing. Data source: Maddison and Gapminder.

Stylized Fact 2: Manufacturing and Education

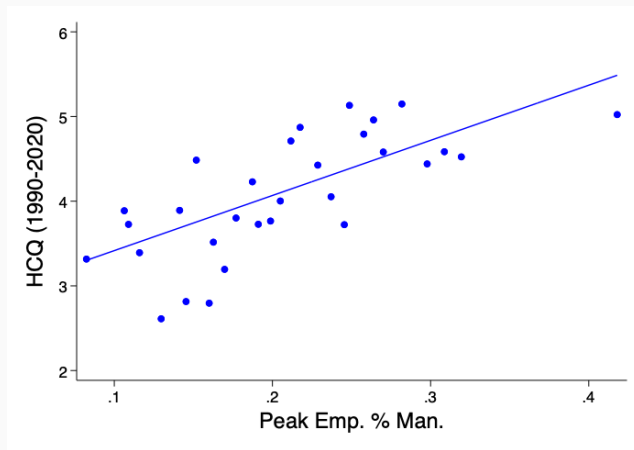
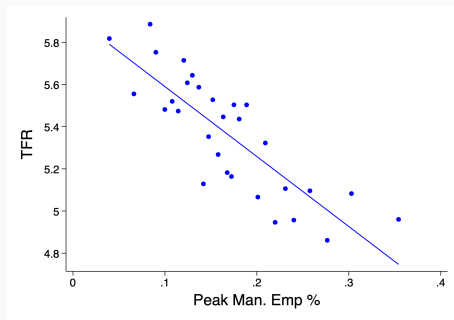


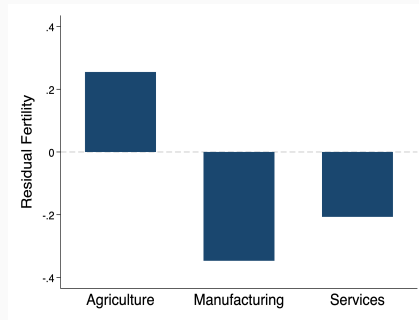
Figure 3: Binned scatterplot of the average Human Capital Quality (HCQ) against the peak manufacturing employment share. Data source: Gapminder and Barro Lee (2015).

► Table

Stylized Fact 3: Fertility and Structural Change



(a) Across countries



(b) Within countries

Figure 4: (left) binned scatterplot of average total fertility rates (TFR) against the peak manufacturing share of employment in 175 countries over 1800–2018 controlling for initial fertility and initial GDP per capita; (right) average residual fertility across sectors within country-year samples of childbearing aged women. Data source: Maddison (2023) and Gapminder (left), IPUMS International (right).

Model

The model is an extension of Galor and Weil (2000) and Lagerlof (2006).

Households:

- **New** $\Rightarrow$ Heterogeneous $i \in \{a, m, s\}$ and live for two periods
- Receive education when young and make labor supply, consumption, fertility and education decisions when old
- QQ tradeoff: the effect of education on human capital depends on the rate of technological progress

Production and dynamics:

- **New** $\Rightarrow$ Representative firms produce agricultural, manufacturing and services goods in a competitive market
- **New** $\Rightarrow$ Technological progress depends on population and government provision of education ρ_t but faces *sector-specific congestion effects* (following Jones, 1995)

Household Problem

Each period households in sector $i \in \{a, m, s\}$ solve the following problem:

$$\tilde{U}_i = \max_{c_{at}, c_{mt}, c_{st}, n_{it}, e_{it}} (1 - \gamma) \ln(C_t - \bar{c}) + \gamma \ln(n_{it} h_{i,t+1}) + \eta_i \quad (1)$$

such that:

$$\begin{aligned} h_{i,t+1} &= \frac{e_{it} + \rho_t \tau_i}{e_{it} + \rho_t \tau_i + g_t} \\ p_{at} c_{at} + c_{mt} + p_{st} c_{st} &\leq (1 - n_{it}(e_{it} + \tau_i)) w_t \\ \sum_{j \in \{a, m, s\}} \omega_j^{\frac{1}{\sigma}} \left(\frac{C_{jt}}{C_t^{\varepsilon_j}} \right)^{\frac{\sigma-1}{\sigma}} &= 1 \end{aligned}$$

With $\gamma, \tau \in (0, 1)$, $\sigma \in (0, 1)$, and p_m normalized to 1 and $\eta_i, \eta_j \in \mathbb{R}^2$ s.t. $\tilde{U}_i = \tilde{U}_j \quad \forall \quad i, j$.

Each sector produces using labor and capital with the following CRS technology:

$$Y_{it} = p_{it} A_{it} K_{it}^{1-\nu} (h_t L_{it})^\nu \quad (2)$$

Where capital is obtained on a world capital market with a fixed interest rate $\bar{r}$ implying:

$$Y_{it} \equiv p_{it} A_{it} (h_t L_{it}) \left(\frac{(1-\nu) p_{it} A_{it}}{\bar{r}} \right)^{\frac{1-\nu}{\nu}}.$$

With $\nu \in (0, 1)$.

Dynamics of Population and Technology

Define $\bar{x} \equiv \sum_i s_i x_i$. Population evolves according to fertility decisions:

$$L_{t+1} = \bar{n}_t L_t \quad (3)$$

Technology evolves in each sector according to:

$$\frac{A_{i,t+1}}{A_{i,t}} = (1 + \underbrace{(\bar{e}_t + \rho_t \bar{r}_t) L_t^{\phi_i}}_{g_{i,t}}) \quad (4)$$

Where:

$$\rho_t = v \left(\frac{\max \{e_\tau\}_{\tau < t}}{b} \right)^\xi$$

Balanced Growth Path and Structural Change

There exists an asymptotic **Balanced Growth Path (BGP)** such that:

1. The employment share of services converges to one:

$$\lim_{t \rightarrow \infty} s_{s,t} = 1,$$

while fertility converges to its replacement level:

$$\lim_{t \rightarrow \infty} n_t = 1.$$

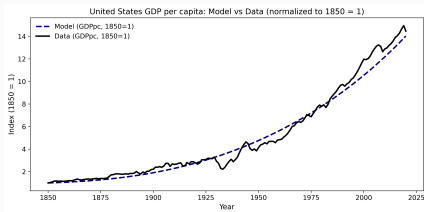
2. Along the BGP, education, growth, and population are constant at:

$$e^* = \gamma - \tau, \quad g^* = \frac{[\gamma + (\rho - 1)\tau]^2}{(1 - \rho)\tau}, \quad L^* = \left[\frac{\gamma + (\rho - 1)\tau}{(1 - \rho)\tau} \right]^{1/\phi_s}.$$

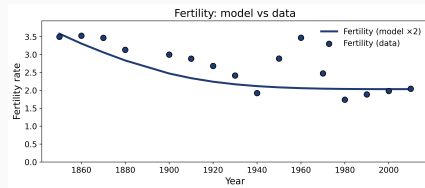
3. The long-run growth rate g^* is **increasing in ρ** .

Calibration

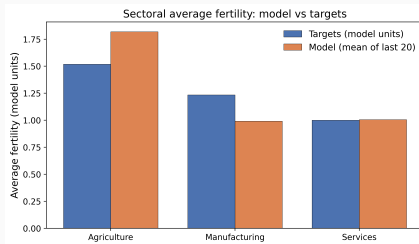
Calibration: USA 1850-2010 GDPpc and Fertility



GDPpc

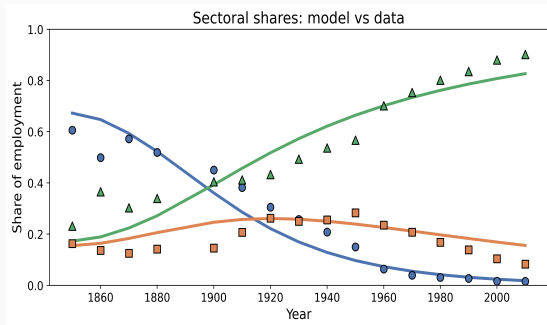


Fertility (NRR)

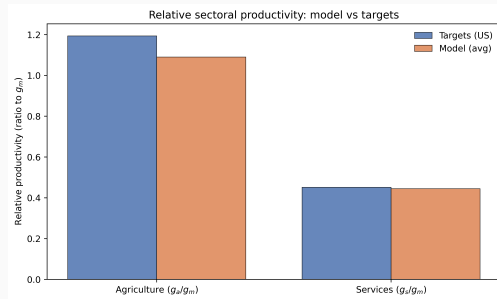


Sectoral Fertility (last 20 years)

Calibration: USA 1850-2010 Sectoral Shares and Productivity



Sectoral shares



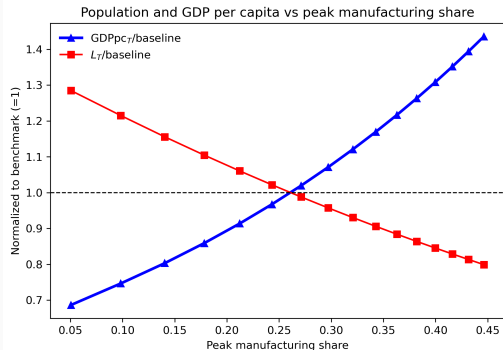
Relative productivity

Results

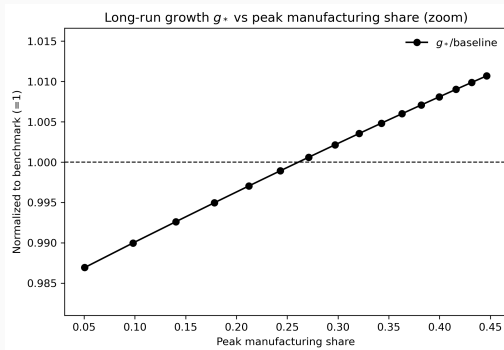
Questions we can ask the model:

- Can we reduce the US' peak manufacturing employment share? What happens to its growth rate, population and GDPPC today?
- How does the model account for the cross-country facts we observe in the data? These are
 - Negative relationship between a country's peak manufacturing employment share and its growth rate.
 - Higher population growth in economies which do not go through manufacturing.
 - Higher levels of education quality (ρ) in countries with higher peaks of manufacturing activity

Results: GDPPC and Population



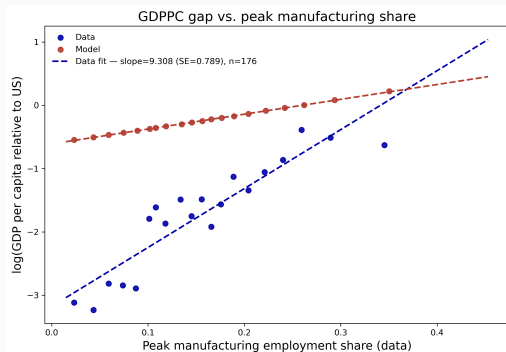
(a) $L_T, GDPPC_T$



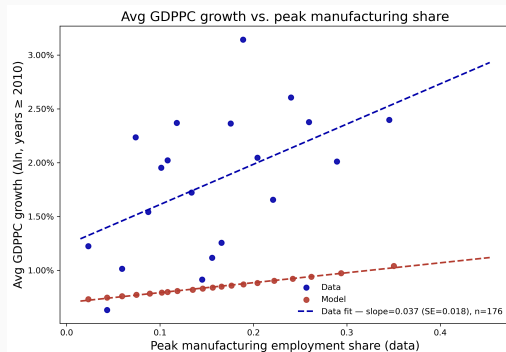
(b) g^*

Figure 5: Final GDPPC and Population for economies with different peak employment shares in manufacturing.

Cross Country Results: GDPPC Gaps and Growth Rates



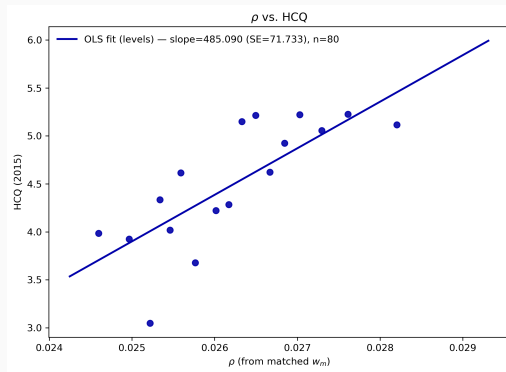
(a) $\log(GDPPC_{c,T}/GDPPC_{USA,T})$



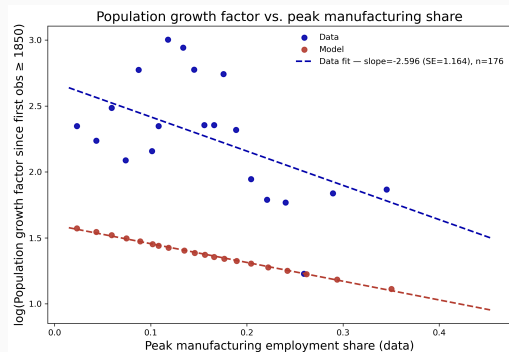
(b) $g_{c,T}^*, g_{c,T}^T$

Figure 6: Final GDPPC and Population for economies with different peak employment shares in manufacturing.

Cross Country Results: Education and Population Growth Factors



(a) ρ , HCQ



(b) $\log(L_T/L_0)$

Figure 7: Binscatter plot of $\ln(GDPPC/GDPPC_{US})_{2018}$ and GDPPC growth rate for economies with different peak employment shares in manufacturing.

Conclusion

Main idea: A growth model with structural change and endogenous fertility explains key global patterns.

- ⇒ Countries with higher manufacturing peaks $\Rightarrow$ faster growth, better education, and quicker demographic transitions.

Model exercise: Calibrate to the US (1850–2010); vary a preference shifter to simulate economies with more/less manufacturing.

Results: The model explains:

- 23.8% of GDP per capita **level gaps**
- 27.2% of GDP per capita **growth gaps**
- 53.9% of population **growth gaps**

Policy

Policy Experiments: Motivation and Overview

- I study how small, budget-balanced policy interventions— either **education subsidies/taxes** or **fertility subsidies/taxes**— affect long-run structural transformation and aggregate welfare.
- Policies are introduced at date T_{pol} and are scaled so that at the moment of implementation they cost a fixed fraction α of the representative household's wage:

$$\frac{B_{T_{\text{pol}}}}{w_{T_{\text{pol}}}} = \alpha.$$

- This allows a clean comparison of policies across instruments:
 - fertility subsidy/tax (s_n)
 - education subsidy/tax (s_e)

All policies have the same fiscal footprint on impact.

Policy Implementation in the Household Problem

Household budget constraint with subsidies:

$$\tilde{C}_i = w_t(1 - n_i(e_i + \tau_i)) + s_e e_i + s_n n_i,$$

where s_e is an education subsidy (or tax) and s_n a fertility subsidy (or tax).

Government budget: in every period the government balances its budget through a lump-sum tax B_t :

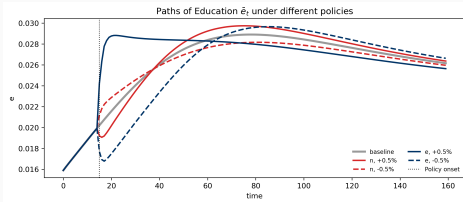
$$B_t = \sum_{i \in \{a, m, s\}} s_i(t) [s_e e_i(t) + s_n n_i(t)],$$

Actual consumption used in demand and welfare:

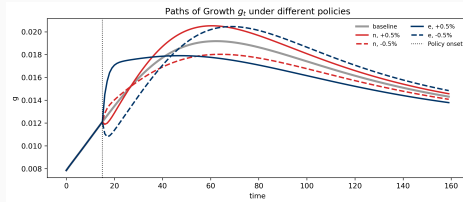
$$C_i(t) = \tilde{C}_i(t) - B_t.$$

The household does not internalize the effect that changing its decisions has on the lump-sum transfer $\partial B_t / \partial x_t = 0$ for all choice variables x_t .

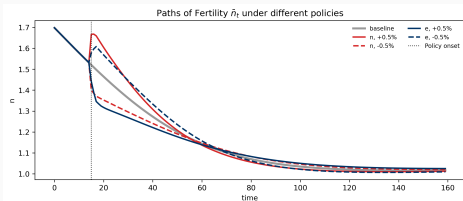
Policy Counterfactuals: Paths



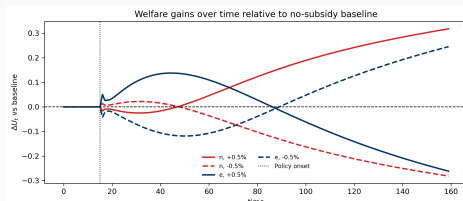
Education



Growth



Fertility



Welfare

Policy Counterfactuals: CEV

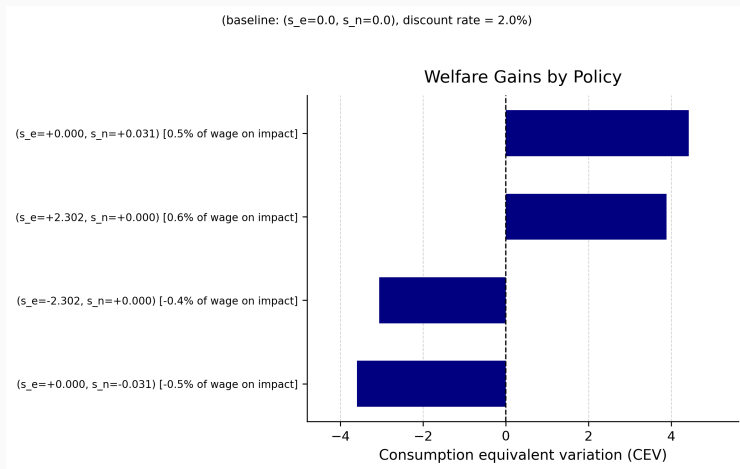
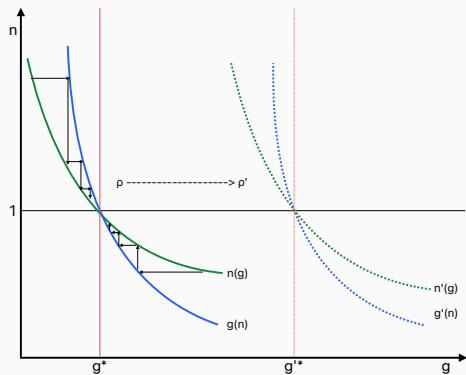


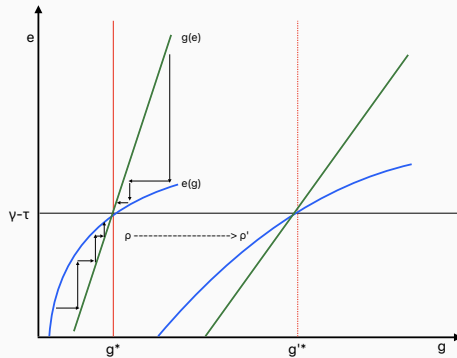
Figure 8: Ranking of policies by CEV

appendix

Model: Phase Diagram



(a) e and g



(b) n and g

Fact 1: Fertility and Structural Change

Table 1: OLS Regressions

	(1)	(2)	(3)	(4)	(5)
% Man. Emp.	-13.43*** (-10.74)	-4.815*** (-4.85)	-4.815*** (-4.85)	-6.113*** (-5.40)	-6.826*** (-5.54)
Log GDP p.c.		-0.808*** (-5.58)	-0.808*** (-5.58)	-0.782*** (-5.33)	0.355** (3.29)
% Agr.		0.809 (1.07)	0.809 (1.07)	0.559 (0.72)	0.0945 (0.12)
N	5053	5053	5053	5053	5053
R2	0.326	0.655	0.655	0.676	0.966
Country FE	no	no	no	no	yes
Year FE	no	no	no	yes	yes

t statistics in parentheses

Dependent variable is total fertility rate. Standard errors are clustered at the country level.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Empirical Analysis

Crosswalk creation

- Very few national statistical agencies classify according to a harmonized industrial standard (e.g. ISIC).
- In practice classifications vary not only by country but also by year. In 44 samples (country-year pairs) we observe over 5000 *different* industries, e.g.:

Sample	Industry Description	SITC4 Code
Ghana 2000	Agriculture and animal farming combined	0111, 0112, 0113,...
Ghana 2010	Agriculture and hunting	0111, 0112, 0113,...
Ethiopia 1984	Agriculture and livestock production	0111, 0112, 0113,...

- The crosswalk is created using the ChatGPT API.

Empirical Methods

Using this crosswalk, we can assess how trade shocks affect regional outcomes in all the countries covered by IPUMS International. We do this with the following regression model:

$$y_{i,c,>2000} - y_{i,c,1980-2000} = \alpha_c + \beta \times \ln(\text{shock}_{i,c}) + \epsilon_{i,c} \quad (5)$$

Where y is a regional level outcome, α is a country FE and:

$$\text{shock}_{i,c} = \sum_j \bar{s}_{i,c,j} (M_{j,1990-2000} - M_{j,1975-1990})$$

With $\bar{s}_{i,c,j}$ is the average share of employment in industry j , region i , country c over the entire sample.

Empirical Results

Table 2: OLS Regressions

	(1)	(2)	(3)	(4)	(5)
	Manufacturing	Serv. High	Serv. Low	Agriculture	Fertility
(mean) ln_shock	-0.0167* (-2.30)	-0.000955 (-0.66)	0.0225** (2.73)	-0.00671 (-1.13)	0.0117 (0.29)
N	460	460	460	460	448
R2	0.692	0.828	0.539	0.577	0.663
Country FE	yes	yes	yes	yes	yes

t statistics in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Stylized Fact 1: Fertility and Structural Change

To quantify differences in fertility across sectors I run the following regression on a sample of 46 million women between the ages of 15 and 45 from 89 low and middle income countries:

$$nchild_{i,c,t} = \alpha_{a,c,t} + \delta_{e,c,t} + \gamma_{n,c,t} + \xi_{u,c,t} + \varepsilon_{i,c,t} \quad (6)$$

Where $nchild_{i,c,t}$ is the number of children for a survey respondent in country c and year t , α is a country-year-age fixed effect, δ is a country-year-education fixed effect, γ is a country-year-non-working fixed effect and ξ is a country-year-urban fixed effect.

Stylized Fact 1: Fertility and Structural Change

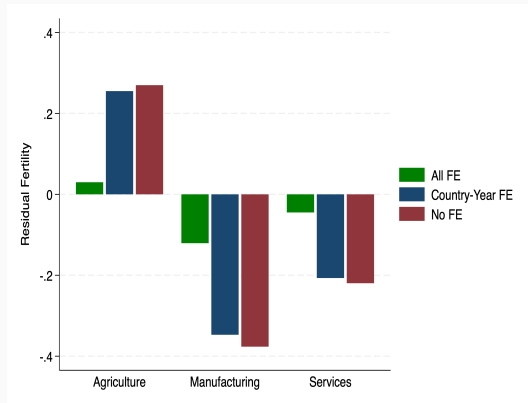


Figure 9: Average residual by sector from model 6 for different FE models. Data Source: IPUMS International.

Regression Table: GDP p.c. Growth rate in 2010-2020

Table 3: Dep. Var. is Avg GDPPC growth rate 2010-2020

	(1)	(2)	(3)	(4)	(5)
Peak Emp. % Man.	5.196** (2.01)	2.478 (0.95)	5.673* (1.77)	7.116* (1.90)	4.093 (1.21)
% Serv. Emp.	-7.243*** (-4.58)	-8.817*** (-5.09)	-6.931** (-2.13)	-5.394 (-1.37)	-6.433* (-1.87)
Human Capital Quality (2015)		0.531** (2.19)			0.850** (2.50)
Log. GDP per capita			-0.00525 (-0.01)	1.412 (1.33)	1.160 (1.21)
N	80	80	80	80	80
R2	0.228	0.268	0.230	0.304	0.362
Income Decile FE	no	no	no	yes	yes
Region FE	no	no	no	yes	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Regression Table: GDP p.c. Growth rate in 2010-2020 Full Sample

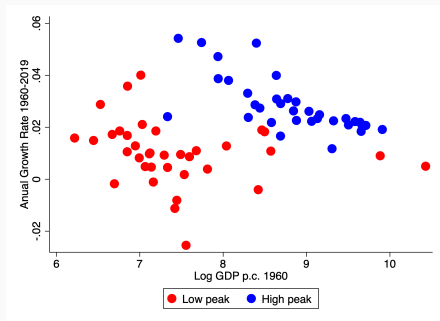
Table 4: Dep. Var. is Avg GDPPC growth rate in 2010-2019

	(1)	(2)	(3)	(4)	(5)
Peak Emp. % Man.	3.738** (2.06)	5.710** (2.38)	5.363** (2.00)	5.624** (2.24)	4.825* (1.71)
% Serv. Emp.		-7.694*** (-3.10)	-7.673*** (-2.76)	-7.885*** (-3.14)	-7.433*** (-2.69)
Log. GDP per capita		0.893** (2.12)	0.731 (1.26)	2.093** (2.06)	1.956* (1.78)
N	176	176	176	176	176
R2	0.0176	0.155	0.169	0.208	0.220
Income Decile FE	no	no	no	yes	yes
Region FE	no	no	yes	no	yes

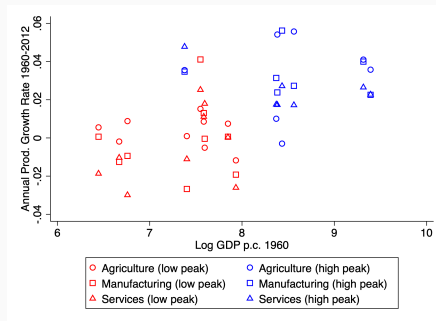
t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 3: Lucas 2009 Style Plot



(a) Aggregate p.c. Growth



(b) Sector Decomposition

Figure 10: Scatterplot of average annual growth of GDP per capita over 1960-2020, distinguishing between high (blue) and low (red) peak manufacturing employment economies. Data source: Gapminder (left) and GGDC 10-sector (right).

Stylized Fact 3: Decomposition Across Sectors

Table 5: Decomposition of Prod. Gr. Effects

	(1)	(2)	(3)	(4)	(5)
	Agr.	Man.	Serv.	FIRE	Non-FIRE
Peak Emp. % Man.	-0.0129 (-0.29)	0.0699** (2.25)	0.0220 (0.49)	0.133** (2.60)	0.0140 (0.27)
N	41	41	41	41	41
R2	0.00358	0.108	0.00887	0.146	0.00313

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 2: Education Table 1/2

Table 6: Dependent Variable is avg. HCQ (1990-2020)

	(1)	(2)	(3)	(4)
Peak Emp. % Man.	6.512*** (6.34)	6.971*** (5.09)	2.055 (1.56)	1.990* (1.91)
% Man. Emp.		-1.461 (-0.74)	2.706* (1.83)	0.662 (0.43)
N	80	80	80	80
R2	0.303	0.308	0.569	0.695
Region FE	no	yes	no	yes
Controls	no	no	yes	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 2: Education Table 2/2

Table 7: Dependent Variable is avg. TFR (1990-2020)

	(1)	(2)	(3)	(4)
HCQ (1990-2020)	-0.579*** (-8.06)	-0.449*** (-5.71)	-0.367*** (-4.27)	-0.216* (-1.91)
Peak Emp. % Man.		-2.810*** (-4.07)	-3.222*** (-4.31)	-2.962*** (-3.61)
N	80	80	80	80
R2	0.503	0.561	0.604	0.665
Region FE	no	no	no	yes
Controls	no	no	yes	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 3: Heterogeneity in the Service Sector

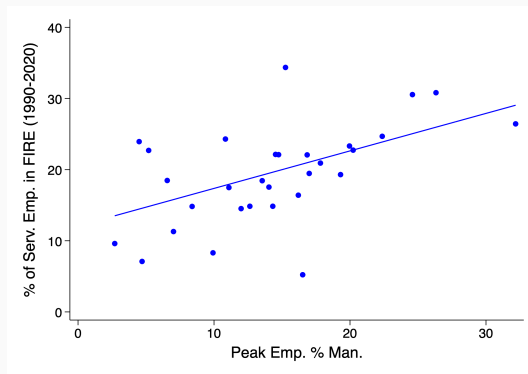
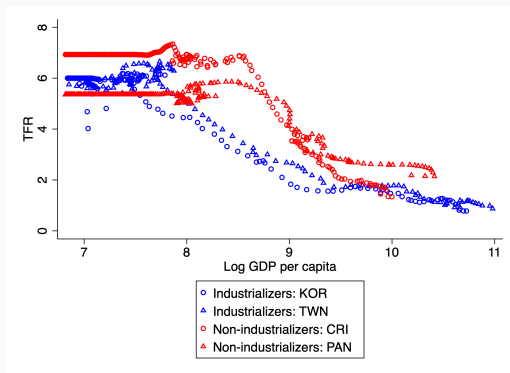
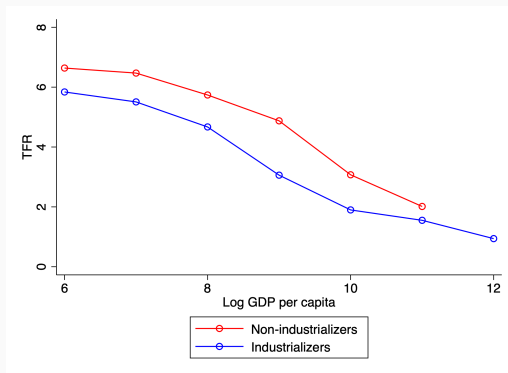


Figure 11: Binscatter plot of the fraction of service workers working in high productivity services (FIRE plus health-care and legal) against the peak employment share in manufacturing. Data source: IPUMS International for the fraction of high productivity services, GGDC for the peak employment shares of manufacturing.

Stylized Fact 1: Fertility and Structural Change



(a) Country-Comparison



(b) q1 vs q5

Figure 12: Scatterplot of the Total Fertility Rate (TFR) over log of GDP per capita in four countries. Red dots correspond to major industrializers and blue to non-industrializers. Data Source: Gapminder.

Table 8: Dep. Var. is Avg GDPPC growth rate 2010-2020

	(1)	(2)	(3)	(4)	(5)
Peak Emp. % Man.	5.196** (2.01)	2.478 (0.95)	5.673* (1.77)	7.116* (1.90)	4.093 (1.21)
% Serv. Emp.	-7.243*** (-4.58)	-8.817*** (-5.09)	-6.931** (-2.13)	-5.394 (-1.37)	-6.433* (-1.87)
Human Capital Quality (2015)		0.531** (2.19)			0.850** (2.50)
Log. GDP per capita			-0.00525 (-0.01)	1.412 (1.33)	1.160 (1.21)
N	80	80	80	80	80
R2	0.228	0.268	0.230	0.304	0.362
Income Decile FE	no	no	no	yes	yes
Region FE	no	no	no	yes	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Competitive Equilibrium

A competitive equilibrium in this economy consists of household decisions c_a, c_m, c_s, n, e, i , firm decisions L_a, L_m, L_s , prices w_a, w_m, w_s, p_m, p_s , and sectoral technologies A_a, A_m, A_s . Such that, given prices:

- **Labor market:** Total labor supply equals total demand:

$$L = \sum_{i \in \{a, m, s\}} L_i.$$

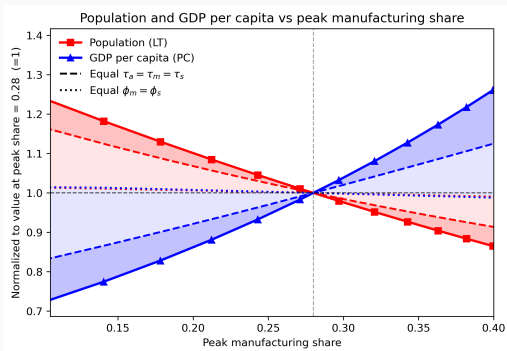
- **Goods markets:** Each sector's output equals demand:

$$y_i = c_i, \quad i \in \{a, m, s\}.$$

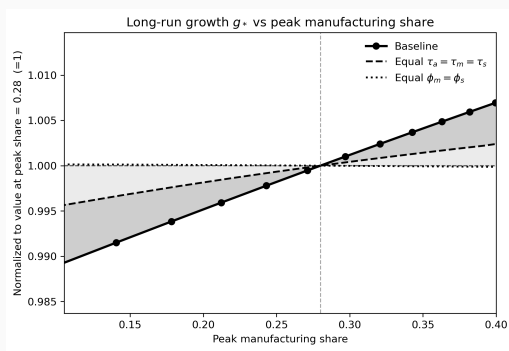
- **Optimization:** The household decisions solve the problem given in 1.

The economy is on a BGP if $g_{i,t} = g_{i,t+1} \forall i$.

Results: GDPPC and Population Decomposition



(a) $L_T, GDPPC_T$



(b) g^*

Figure 13: Final GDPPC and Population for economies with different peak employment shares in manufacturing.

Regression Table: GDP p.c. and Peak % Man. Emp.

Table 9: Dependent var is $\ln(\text{GDPPC})$

	(1)	(2)	(3)	(4)	(5)
Peak Emp. % Man.	9.308*** (11.80)	8.898*** (9.69)	3.182*** (3.91)	4.620*** (4.97)	1.950*** (2.63)
Years Since Peak		0.00595 (1.26)	-0.00136 (-0.36)	0.0103** (2.23)	0.00104 (0.28)
N	176	176	176	176	176
R2	0.426	0.430	0.766	0.605	0.812
Region FE	no	no	no	yes	yes
WB Institutions Controls	no	no	yes	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Fertility and Education Choices

Define $\tilde{w} \equiv \frac{\bar{c}}{1-\gamma}$, the level of w such that if the household allocates $(1 - \gamma)$ of its resources to consumption, it can just satisfy the subsistence constraint. Then for all households i :

$$n_{it}(\tau_i + e_{it}) = \begin{cases} \gamma & \text{if } w_t \geq \tilde{w}, \\ 1 - \frac{\bar{c}}{w_t} & \text{if } w_t \in (\bar{c}, \tilde{w}), \\ 0 & \text{if } w_t \leq \bar{c} \end{cases} \quad (7)$$

Implications for effective resources:

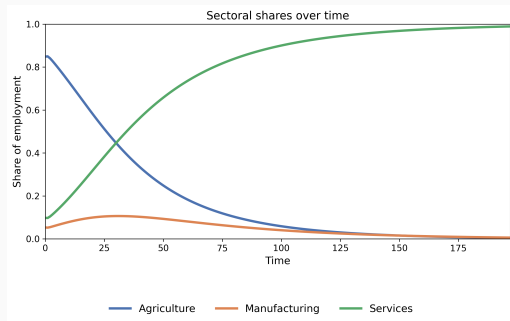
$$\underbrace{(1 - n_{it}(\tau_i + e_{it}))}_{R_t} w_t = \begin{cases} w_t(1 - \gamma) & \text{if } w_t \geq \tilde{w}, \\ \bar{c} & \text{if } w_t \in (\bar{c}, \tilde{w}) \end{cases}$$

Optimal education choice:

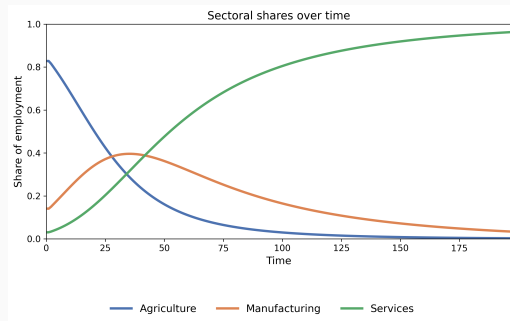
$$e_{it}^* = \max \left\{ 0, \sqrt{\bar{g}_t(1 - \rho_t)\tau_i} - \rho_t\tau_i \right\}$$

Model: Qualitative Patterns 1/2

Consider two economies with different ω_m :



(a) $\omega_m = 0.5$



(b) $\omega_m = 3.5$

Model: Qualitative Patterns 2/2

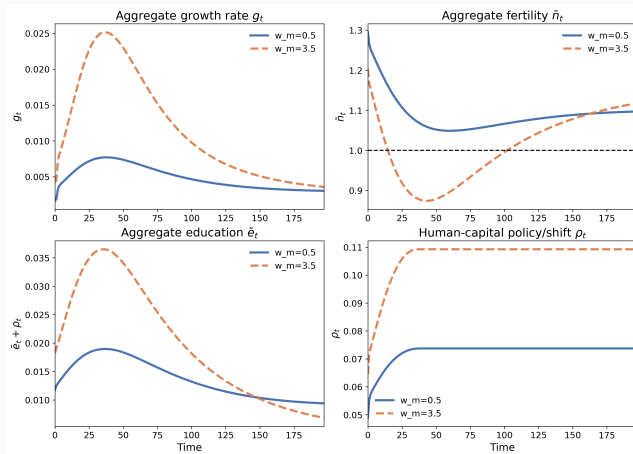


Figure 14: Comparison of Dynamics for two economies. The blue line corresponds to the economy which does NOT go through manufacturing (low peak), and the orange line the economy which does go through manufacturing (high peak).